Quantum Gravitation

Lecture 7

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KAIST

Source Coupling and Scattering Amplitudes

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$\mathcal{L}_{\text{int}} = -(\kappa/2)h_{\mu\nu}(T^{\mu\nu} + S^{\mu\nu})$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$\mathcal{M} \sim \kappa^2 T_1 \Pi^{(2)} T_2 / (q^2 + m_g^2)$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$V(r) = -Gm_1 m_2 e^{-m_{\mathrm{g}} cr/\hbar}/r$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Seo, ch. 7; Scattering Reference SL-66; Cole, ch. 10